

$$\text{Final velocity} = -1.0 + 0.8 = -0.2 \text{ m s}^{-1}$$

4

D

At maximum height, $v_y = 0$ while $v_x = v \cos \theta$
 $\therefore \text{K.E. at maximum height} = \frac{m(v \cos \theta)^2}{2}$
 $= \text{K.E.}_{\text{initial}} \cos^2 \theta$
 $= 15 \cos^2 30^\circ$
 $= 11 \text{ J (2sf)}$